

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0423

Barling 345 kV
September 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0423. The Interconnection Customer proposes a 599.2 MW wind facility interconnecting at the Barling 345 kV substation in the Maryland area. The request was received on January 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 206 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$594,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Irvine / Marcus Hook Energy Lp Substation 500kV
Contingency: Guilbeau Electric / Marcus Hook Energy Lp 500kV
Loading: 102.3% (Rating: 2,306 MVA)
DFAX: 7.2%
Pre-project loading: 100.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Irvine / Marcus Hook Energy Lp Substation 500kV	Reconductor line	\$11,396,000	5% (\$594,000)
	Total allocated to project		\$594,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.